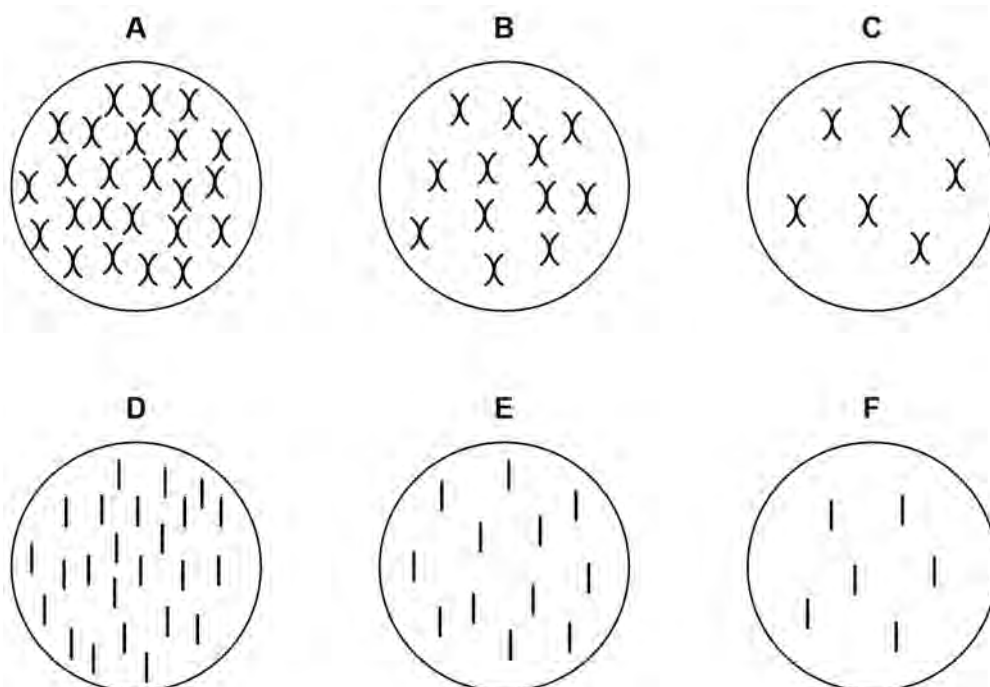


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Figure 3 shows diagrams of six possible arrangements of chromosomes in cells.

Do not write
outside the
box

Figure 3



Before meiosis, a cell of a rice plant has 12 pairs of homologous chromosomes (24 chromosomes in total).

Give the letter of the diagram from **Figure 3** that correctly shows the chromosome content of rice cells after the first meiotic division and after the second meiotic division.

[2 marks]

After first meiotic division _____

After second meiotic division _____



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Scientists have produced a mutated rice variety in which there is **no** crossing over.

A population of the mutant rice variety produced by sexual reproduction shows genetic variation. Populations of non-mutant rice varieties also show genetic variation.

Suggest and explain the similarities and differences in the causes of genetic variation within these rice populations.

[3 marks]

5

Turn over for the next question

Turn over ►

